



Zero to Hero
Series



Learn.
Practice.
Defend.



MICROSOFT SENTINEL

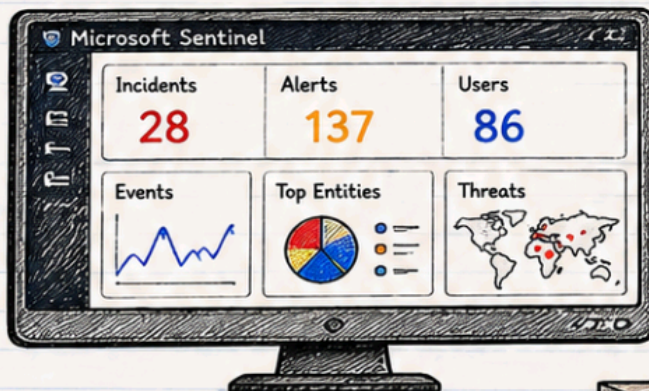
ZERO TO HERO

NOTEBOOK

The Beginner-Friendly SOC Analyst Guide



- ✓ Collect. Detect. Investigate.
- ✓ Respond. Automate.
- ✓ All in the Cloud.
- ✓ Built for the SOC.
- ✓ Powerful. Scalable. Smart.
- ✓ Made for Analysts like You.



Detect threats.
Investigate incidents.
Protect your organization.



★ This notebook is your complete roadmap to learn Microsoft Sentinel from the basics to advanced SOC analyst skills.



Collect
Logs



Detect
Threats



Investigate
Incidents



Automate
Response



Visualize
& Report

Perfect for:

- Beginners
- SOC Analysts
- Blue Teamers
- Cybersecurity Learners



You don't need to be an expert to start.
You just need to start to become an expert.



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1. WHAT IS MICROSOFT SENTINEL?



"What if your SIEM lived in the cloud?"



① WHAT IS MICROSOFT SENTINEL?

Microsoft Sentinel is a cloud-native SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation and Response) solution built on Microsoft Azure.

In simple words:

Sentinel collects data, detects threats, investigates incidents and helps you respond - all from the cloud.

② WHAT IS SIEM?

SIEM stands for Security Information and Event Management.

A SIEM tool helps organizations to:

- ✓ Collect and centralize logs
- ✓ Monitor systems and users
- ✓ Detect suspicious activities
- ✓ Investigate security incidents
- ✓ Generate alerts and reports
- ✓ Improve overall security



③ WHY MICROSOFT CREATED SENTINEL?



To provide a modern, cloud-first SIEM for today's cloud environments.



To help organizations reduce cost and complexity of traditional SIEM solutions.



To deliver scalable, intelligent and automated security operations.



To give security teams better visibility across the entire environment.

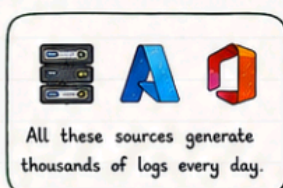
④ SENTINEL vs TRADITIONAL SIEM

Feature	Traditional SIEM	Microsoft Sentinel
Deployment	On-Premises	Cloud (Azure)
Scalability	Limited (Hardware)	Highly Scalable
Maintenance	High (Servers, Updates)	Low (Managed by Microsoft)
Cost	High (Infra + Licenses)	Pay-as-you-go
Integration	Limited	Deep Azure Integration
Automation	Limited	Built-in SOAR
Accessibility	Office Network	Anywhere (Cloud)

⑤ REAL-WORLD EXAMPLE

Imagine a company that uses:

- Windows Servers
- Azure Cloud Services
- Microsoft 365
- Firewalls
- Applications



KEY TAKEAWAY:

Microsoft Sentinel is a modern, cloud SIEM built for today's security challenges.

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2. WHY SOC ANALYSTS USE SENTINEL



"Millions of logs. One place to investigate."

Microsoft Sentinel helps SOC Analysts collect data, detect threats, investigate incidents and respond faster - all in the cloud.



REAL-WORLD EXAMPLE

A company has:

- 100+ Windows servers
- Azure resources
- Microsoft 365 users
- Firewalls, VPNs, Apps



→ All these systems generate thousands of logs every day.

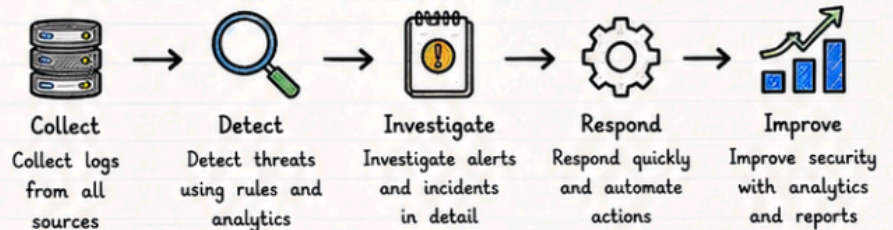
Microsoft Sentinel collects all logs in one place, detects suspicious activity and helps analysts respond quickly.

KEY BENEFITS

- ✓ Centralized visibility across your entire environment
- ✓ Real-time threat detection and alerts
- ✓ Built-in threat intelligence
- ✓ Powerful hunting with KQL
- ✓ Automation and SOAR capabilities
- ✓ Scalable and cost-effective cloud solution
- ✓ Native integration with Microsoft ecosystem



HOW IT HELPS SOC ANALYSTS



SENTINEL IN ACTION (REAL PORTAL SCREENSHOT)



WHY IT MATTERS

- ★ See everything in one place.
- ★ Detect threats faster.
- ★ Investigate with powerful tools.
- ★ Automate repetitive tasks.
- ★ Make data-driven security decisions.



KEY TAKEAWAY:

Microsoft Sentinel gives SOC Analysts the visibility, tools and automation they need to protect their organization in today's cloud world.



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3. HOW MICROSOFT SENTINEL WORKS



"Every alert starts with data."

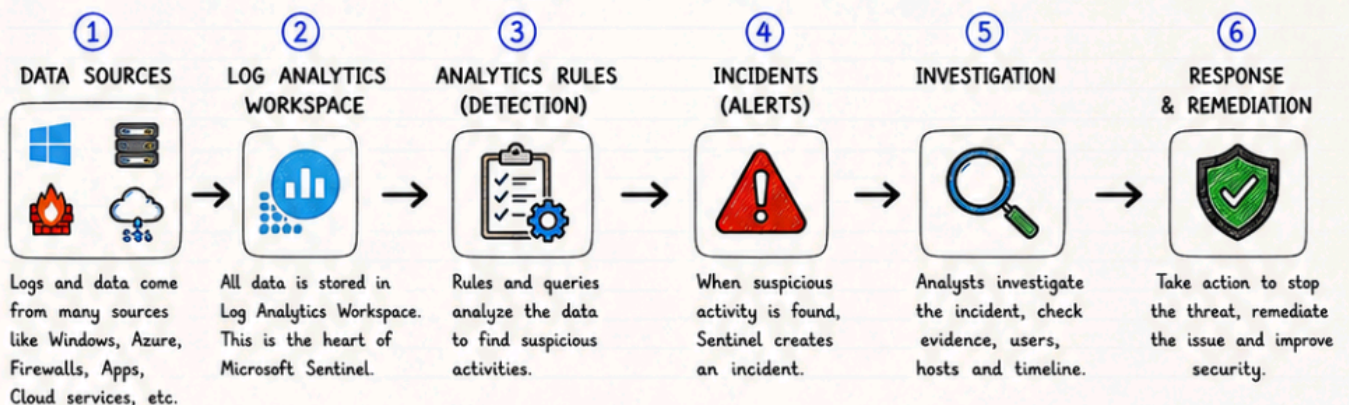
Microsoft Sentinel follows a simple but powerful flow. It collects data from many sources, analyzes it and helps SOC Analysts **detect**, **investigate** and **respond** to threats.

Think of it like this:

1. Data comes in
2. Sentinel looks for bad things
3. It creates incidents
4. You investigate and respond



THE MICROSOFT SENTINEL WORKFLOW



KEY POINTS

- ✓ Data is the foundation.
- ✓ Analytics rules are the brain.
- ✓ Incidents are created when something is suspicious.
- ✓ Analysts investigate and respond.
- ✓ Automation helps to respond faster.



REMEMBER:

Microsoft Sentinel is not just a tool, it is a complete cloud platform that helps you **see**, **detect**, **investigate** and **respond** to threats in real-time.

MINI LAB

- 1 Understand the workflow above.
- 2 Think about your environment.
- 3 List 3 data sources you will connect.



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4. MICROSOFT SENTINEL ARCHIECTURE

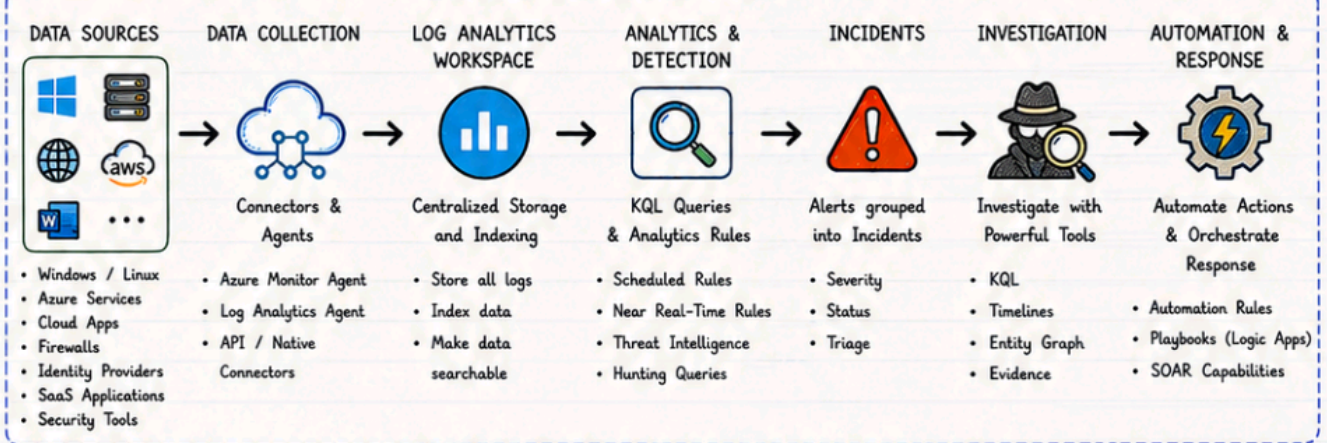


"Know the building before you explore the rooms."



Microsoft Sentinel is built on Microsoft Azure. It uses several Azure services together to collect, store, analyze and respond to security data at scale.

SENTINEL ARCHITECTURE OVERVIEW



SENTINEL ARCHITECTURE IN AZURE PORTAL



KEY COMPONENTS EXPLAINED

- 1 **Data Sources** - Anything that can generate logs.
- 2 **Log Analytics Workspace** - Stores and indexes all data.
- 3 **Analytics & Detection** - Uses KQL and rules to find threats.
- 4 **Incidents** - Alerts are grouped into incidents for investigation.
- 5 **Investigation** - Analysts dig deep with powerful investigation tools.
- 6 **Automation** - Automate tasks and respond faster.

ANALYST TIP

Understanding the architecture helps you know where data comes from, how it flows and where you investigate.

REMEMBER

Sentinel is 100% cloud-native and designed for modern, scalable and cost-effective security operations.

MINI LAB

- 1 Open your Azure Portal.
- 2 Search for Microsoft Sentinel.
- 3 Explore the Overview page.
- 4 Identify the key components.

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5. UNDERSTANDING LOG ANALYTICS WORKSPACE



"The heart of Microsoft Sentinel."

A Log Analytics Workspace is an Azure service that **collects**, **stores** and **analyzes** log data from your environment. It is the **foundation** of Microsoft Sentinel.

Think of it like:

A giant warehouse where all your logs are stored safely.



Microsoft Sentinel

WHY IT IS IMPORTANT?

- ✓ All Sentinel data is stored here.
- ✓ All analytics rules run on data from here.
- ✓ You can run KQL queries on this data.
- ✓ Data is retained based on your settings.
- ✓ It is secure, scalable and highly available.

WHAT IT STORES?

- Security logs (Windows, Linux, Firewalls)
- Azure Activity logs
- Authentication logs
- Application logs
- Custom logs
- Threat intelligence logs and more



REAL PORTAL SCREENSHOT

The screenshot shows the Azure portal interface for a Log Analytics workspace named 'law-sentinel-demo'. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Settings, Locks, Agents management, Data sources, Data storage, Linked storage accounts, Network, and Advanced settings. The main content area displays the 'Essentials' section with various metrics and settings. The 'Usage' section shows 'Ingested data' as 5.6 GB and 'Queried data' as 1.2 GB. The 'Data sources' section shows 230 connected sources. The 'Workspace summary' section shows 30 days of data retention. The 'Tags' section shows 'Environment: Lab'.

KEY POINTS

- ★ All data in Sentinel lives in a Log Analytics Workspace.
- ★ Choose the right region for your data.
- ★ Set proper retention based on your needs and cost.
- ★ More data = Better visibility (and more cost).

RETENTION EXPLAINED

Retention is how long your data is kept.

Example:

- 30 days - Good for small orgs
- 90 days - Good for most SOC's
- 365+ days - Good for investigations



COST TIP

You pay for:

- ✓ Data ingestion (GB/day)
 - ✓ Data retention (days)
 - ✓ Data queries (GB scanned)
- More data and longer retention = higher cost.



MINI LAB

- 1 Go to Azure Portal.
- 2 Create a new Log Analytics Workspace.
- 3 Note the Workspace ID.
- 4 Explore the Overview page.
- 5 Check the Usage and Retention.



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6. INSTALLING MICROSOFT SENTINEL



"Let's build your cloud SOC."

Microsoft Sentinel is deployed on Microsoft Azure. We first create a **Log Analytics Workspace** and then **enable Microsoft Sentinel** on top of it.

What you need:

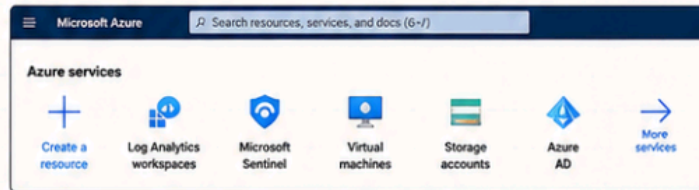
- ☒ Azure Account
- ☒ Subscription with Owner permissions
- ☒ Basic understanding of Azure Portal



STEP-BY-STEP INSTALLATION

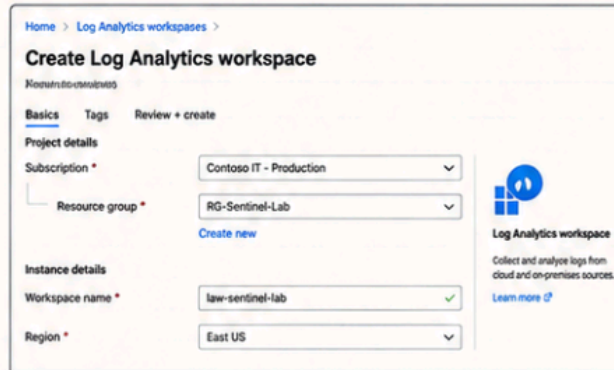
① SIGN IN TO AZURE PORTAL

- Go to portal.azure.com
- Sign in with your account.



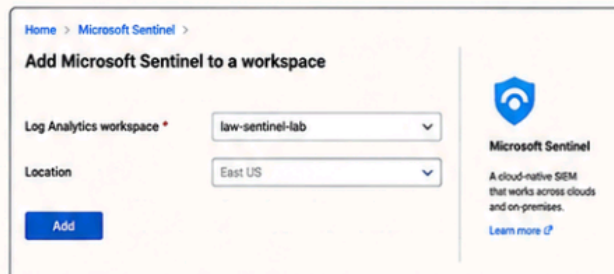
② CREATE LOG ANALYTICS WORKSPACE

- In the search bar, type "Log Analytics workspace" and select it.
- Click on "+ Create".
- Fill in the details:
 - Subscription
 - Resource Group (new or existing)
 - Workspace Name
 - Region
- Click "Review + Create" and then "Create".



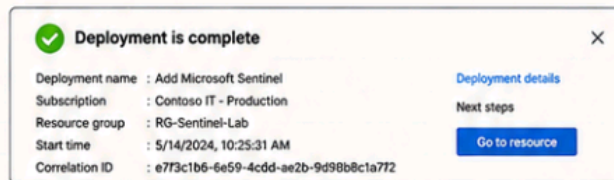
③ ENABLE MICROSOFT SENTINEL

- In the search bar, type "Microsoft Sentinel".
- Select "Microsoft Sentinel".
- Click on "+ Create".
- Select your Log Analytics workspace that you just created.
- Click "Add".



④ DEPLOYMENT IN PROGRESS

- Wait for deployment to complete.
- Once completed, Microsoft Sentinel will be added to your workspace.



KEY TAKEAWAY:

To install Microsoft Sentinel, create a **Log Analytics Workspace** first, then **enable Sentinel** on that workspace.

QUICK TIPS

- ★ Use a meaningful workspace name.
- ★ Choose the right region close to your environment.
- ★ You can add Microsoft Sentinel to only one workspace at a time.
- ★ You can enable Sentinel for existing workspaces too!
- ★ No extra cost for enabling Sentinel, you pay only for ingested data.



MINI LAB

- Open Azure Portal.
- Create a Log Analytics Workspace.
- Enable Microsoft Sentinel on it.
- Go to the Overview page.



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7. FIRST LOGIN TO MICROSOFT SENTINEL



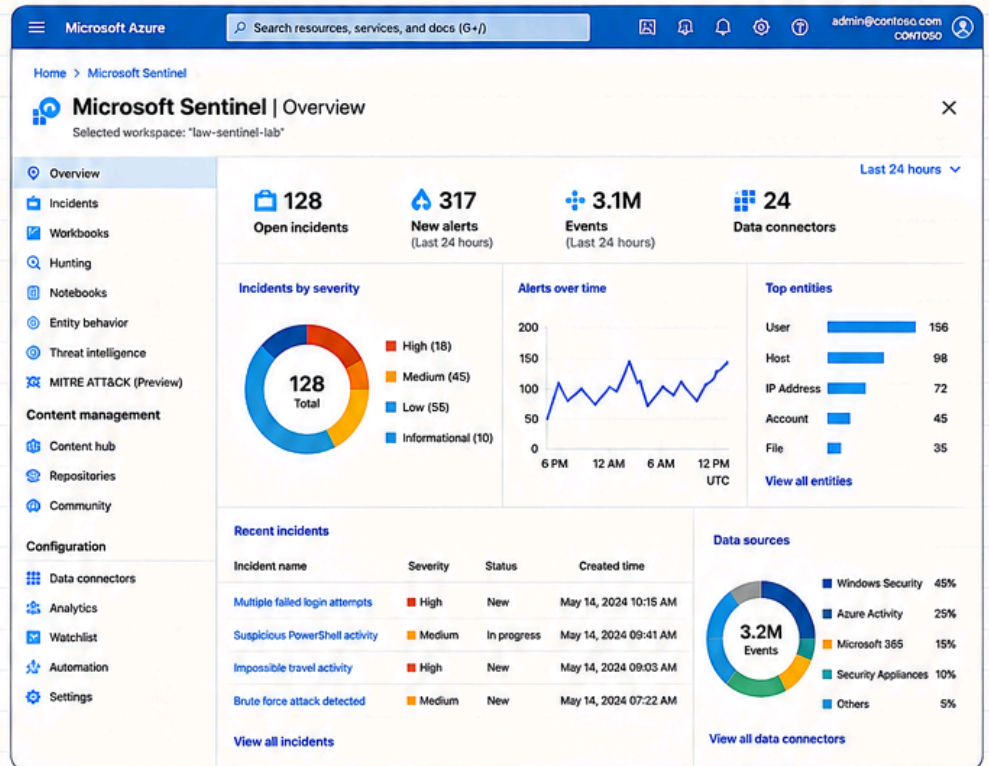
"Welcome to your cloud SOC."



Microsoft
Sentinel

Let's open Microsoft Sentinel for the first time and explore the home page.

- 1 Go to portal.azure.com
- 2 Sign in with your account
- 3 Search "Microsoft Sentinel"
- 4 Select your Log Analytics workspace
- 5 Click "Open"



WHAT YOU SEE ON THE HOME PAGE

- 1 Incidents - Active security incidents that need attention.
- 2 Alerts - New alerts generated by rules and detections.
- 3 Events - Total number of events ingested in the last 24 hours.
- 4 Data connectors - How many data sources are connected.
- 5 Left menu - All main features and tools in Sentinel.



SOC ANALYST TIP



After logging in, always check Open Incidents and New Alerts first. These are your priorities.

REMEMBER



- Sentinel is a cloud SIEM.
- Everything starts with data.
- Data → Alerts → Incidents
- Investigate → Respond

MINI LAB



- 1 Log in to Azure Portal.
- 2 Search "Microsoft Sentinel".
- 3 Open your Sentinel workspace.
- 4 Explore the home page.
- 5 Click on each menu and see what is inside.

NEXT STEP



Now that you are in, let's take a full tour of the Microsoft Sentinel interface.



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8. MICROSOFT SENTINEL INTERFACE TOUR

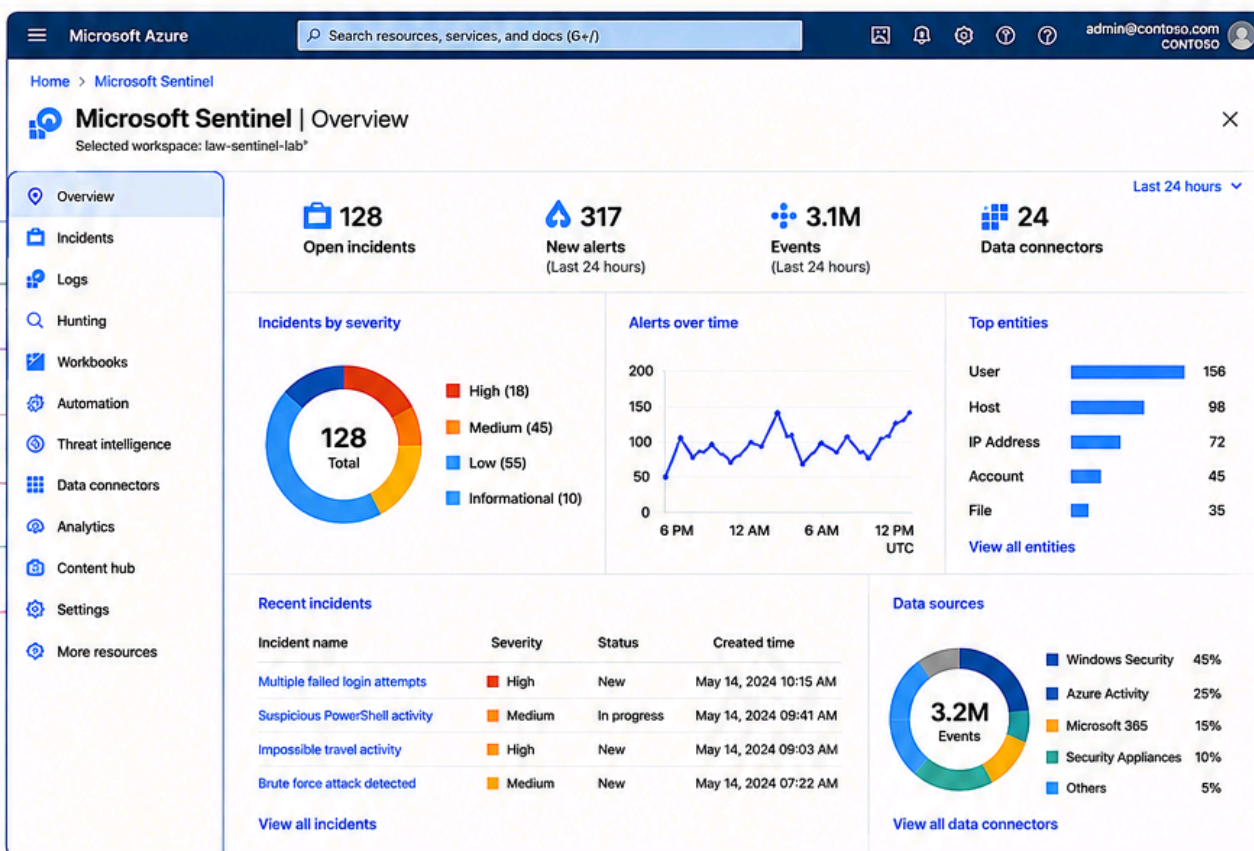


"Know your battlefield before hunting threats."



Microsoft
Sentinel

Microsoft Sentinel has many powerful tools.
Let's understand the main interface.



MAIN MENUS EXPLAINED

1



INCIDENTS

View and manage active security incidents generated by alerts. Investigate, assign, and close incidents.

2



LOGS

Run KQL queries to search and analyze log data from all connected sources.

3



HUNTING

Use advanced hunting queries and notebooks to proactively search for threats.

4



WORKBOOKS

Create interactive dashboards and visual reports to gain visibility into your environment.

5



AUTOMATION

Create automation rules and playbooks to respond to threats automatically.

6



DATA CONNECTORS

Connect data sources like Windows, M365, Azure, and 3rd party tools.

7



CONTENT HUB

Browse and install Microsoft and community content like analytics rules, workbooks, and parsers.



SETTINGS & MORE

Manage workspace settings, roles, data retention, licensing and more configurations.

SOC ANALYST TIP



Spend time exploring each menu. The more you know the interface, the faster you can investigate and respond.

REMEMBER



- Incidents = Respond
- Logs = Search & Analyze
- Hunting = Proactive Search
- Automation = Save Time
- Connectors = Bring in Data

MINI LAB



- Log in to Azure Portal.
- Open Microsoft Sentinel.
- Click on each menu (left side).
- Write down what each one does.
- Take a screenshot of the home page.

NEXT UP



Now that you know the interface, let's connect your data sources.



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9. CONNECTING DATA SOURCES



"No logs = No detections."



Microsoft
Sentinel

Microsoft Sentinel is powerful only when it receives data. Connect data sources to **collect logs** and give Sentinel the visibility it needs to detect threats.

Remember:

More data = Better detections
Better detections = Stronger SOC



STEP-BY-STEP: CONNECT A DATA SOURCE

The screenshot shows the Microsoft Sentinel 'Data connectors' page. The left sidebar contains navigation links: Overview, Logs, News & guides, Threat management (Incidents, Workbooks, Hunting, Notebooks, Entity behavior, Threat intelligence), Content management (Content hub, Repositories, Community), and Configuration (Data connectors, Analytics, Watchlist, Automation, Settings). The main area displays a table of connectors with columns: Connector, Status, Data Types, and Last data received. The table lists 12 connectors, with 10 connected and 2 not connected. The 'Not connected' connectors are Palo Alto Networks and Amazon Web Services.

Connector	Status	Data Types	Last data received
Windows Security Events	Connected	SecurityEvents	2 minutes ago
Syslog	Connected	Syslog	3 minutes ago
Microsoft 365 Defender	Connected	SecurityAlert, DeviceAlert	5 minutes ago
Azure Activity	Connected	AzureActivity	1 minute ago
Microsoft 365	Connected	AuditLogs, SigninLogs	2 minutes ago
Office 365	Connected	OfficeActivity	2 minutes ago
Azure AD	Connected	SigninLogs, AuditLogs	3 minutes ago
Linux Syslog (CEF)	Connected	Syslog	6 minutes ago
Palo Alto Networks	Not connected	CommonSecurityLog	-
Amazon Web Services	Not connected	AWSCloudTrail	-

STEPS

- 1 Go to **Data connectors** (left menu).
- 2 Click on **" + Add connector "**.
- 3 Choose the data source you want to connect. (e.g., Windows, M365, Azure AD)
- 4 Follow the setup wizard.
- 5 Validate that the status shows **Connected**.
- 6 Start receiving logs! ✓

COMMON DATA SOURCES & WHAT THEY PROVIDE

WINDOWS SECURITY

- Security Event Logs
- Logon/Logoff
- Process Creation
- File Modifications

LINUX SYSLOG

- Auth Logs
- Sudo Logs
- Syslog Events
- App Logs

MICROSOFT 365

- Sign-in Logs
- Audit Logs
- User Activities
- Email & Threat Protection

AZURE ACTIVITY

- Resource Changes
- Role Assignments
- Azure Services Operations
- Policy Events

MICROSOFT DEFENDER

- Security Alerts
- Device Alerts
- Threat Detections
- Vulnerabilities

SOC ANALYST TIP

Connect as many relevant sources as possible. The more data you collect, the faster you can detect real threats.

REMEMBER

- No data = No visibility
- No visibility = No detections
- More data = More insights
- **Validate** after connecting!

MINI LAB

- 1 Go to **Data connectors**.
- 2 Click **" + Add connector "**.
- 3 Add Windows Security Events.
- 4 Check Status = **Connected**.
- 5 Open Logs and run a query to see the data.

WHAT'S NEXT?

Now that data is flowing, let's explore the Data Connectors in detail on the next page.



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10. DATA CONNECTORS EXPLAINED



"Sentinel speaks many security languages."



Microsoft
Sentinel

Data connectors bring log data from different sources into Microsoft Sentinel. Once connected, Sentinel can analyze the data and detect threats.

Connector	Provider	Data Types	Status	Last data received
Windows Security Events	Microsoft	SecurityEvents	Connected	2 minutes ago
Syslog	Microsoft	Syslog	Connected	3 minutes ago
Microsoft 365 Defender	Microsoft	SecurityAlert, DeviceAlert	Connected	5 minutes ago
Azure Activity	Microsoft	AzureActivity	Connected	1 minute ago
Microsoft 365	Microsoft	AuditLogs, SigninLogs	Connected	2 minutes ago
Office 365	Microsoft	OfficeActivity	Connected	2 minutes ago
Azure AD	Microsoft	SigninLogs, AuditLogs	Connected	3 minutes ago
Linux Syslog (CEF)	Microsoft	Syslog	Connected	6 minutes ago
Palo Alto Networks	Palo Alto	CommonSecurityLog	Not connected	-
Amazon Web Services	AWS	AWSCloudTrail	Not connected	-

QUICK TIPS

- ★ Make sure the connector status is **Connected**.
- ★ If logs are not showing, check the **Last data received** time.
- ★ You can connect multiple sources to get full visibility.
- ★ More data sources = More detections = Stronger SOC.



UNDERSTANDING THE COLUMNS

1 CONNECTOR	2 PROVIDER	3 DATA TYPES	4 STATUS	5 LAST DATA RECEIVED
The name of the data source you connect to Sentinel. Examples: Windows Security Events, Azure Activity, Microsoft 365.	The vendor or service provider that supplies the data. Most built-in connectors are from Microsoft.	The type of logs or events that will be sent to Sentinel. Examples: SecurityEvents, Syslog, AuditLogs, SigninLogs.	Whether the connector is connected and sending data successfully. ✓ Connected ✗ Not connected	Shows the last time data was ingested from the source. If this is old or "-", check your connector configuration.

SOC ANALYST TIP

Start with these essential connectors for any SOC:

- ✓ Windows Security Events
- ✓ Microsoft 365 Defender
- ✓ Azure Activity
- ✓ Azure AD (SigninLogs)
- ✓ Syslog / Firewall logs

REMEMBER

- No data connector = No logs.
- No logs = No detection.
- Validate after connecting.
- Keep connectors healthy and monitored.

MINI LAB

- 1 Go to **Data connectors**.
- 2 Click "+ Add connector".
- 3 Choose "Windows Security Events".
- 4 Follow the setup wizard.
- 5 Wait for status to show "Connected".
- 6 Open Logs and run a query to confirm data is coming in.

WHAT'S NEXT?

Great! Now that your data is flowing into Sentinel, let's explore the Logs section and start writing KQL queries.

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11. LOGS EXPLAINED



"Every investigation starts with a log."



Microsoft
Sentinel

Logs are the raw data collected from your connected sources.
Use Logs to search, filter and analyze events with KQL.

The screenshot shows the Microsoft Sentinel Logs interface. On the left, there's a sidebar with 'Log management' and a list of logs including 'SecurityEvent'. The main area shows a KQL query: `SecurityEvent | where TimeGenerated > ago(24h) | take 50`. The results table displays columns: TimeGenerated [UTC], Computer, Account, EventID, LevelDisplayName, and IpAddress. The results show five log entries for the 'SecurityEvent' table, all from 'DC01' computer, with various accounts and EventIDs.

TimeGenerated [UTC]	Computer	Account	EventID	LevelDisplayName	IpAddress
5/14/2024, 10:24:15.123 AM	DC01	CONTOSO\jdoe	4624	Information	10.0.0.15
5/14/2024, 10:24:16.456 AM	DC01	CONTOSO\asmith	4625	Warning	10.0.0.25
5/14/2024, 10:24:17.789 AM	DC01	CONTOSO\jdoe	4634	Information	10.0.0.15
5/14/2024, 10:24:18.912 AM	DC01	CONTOSO\svc_sql	4720	Information	10.0.0.30
5/14/2024, 10:24:20.111 AM	DC01	CONTOSO\jdoe	4672	Information	10.0.0.15

QUICK TIPS

- ★ Start with a time range. Smaller range = faster results.
- ★ Use specific columns to get better insights.
- ★ Use filters to narrow down the data.
- ★ Save useful queries for future investigations.

KEY LOG COLUMNS (USING SECURITYEVENT AS EXAMPLE)

#	COLUMN	WHAT IT MEANS	EXAMPLE	WHY IT MATTERS
1	TimeGenerated	The time the event was recorded in Sentinel (UTC).	5/14/2024, 10:24:15 AM	Helps build timelines and correlate events.
2	Computer	The host or device where the event occurred.	DC01	Identify which system was involved.
3	Account	The user or service account involved in the event.	CONTOSO\jdoe	Understand who did what.
4	EventID	The Windows event ID that describes the action.	4624 (Successful logon)	Key to understand the type of activity.
5	IpAddress	The source IP address of the event.	10.0.0.15	Useful to identify external or suspicious IPs.

REAL LOG EXAMPLES

Successful Logon (4624)

User jdoe logged in successfully from IP 10.0.0.15.

Failed Logon (4625)

User asmith failed to log in from IP 10.0.0.25.

Account Lockout (4740)

User jdoe account was locked due to too many failed logons.

SOC ANALYST TIP

Always validate your findings with multiple logs and time correlation. One log might be normal, but patterns reveal the truth!

REMEMBER

- Logs = Raw data
- Raw data = Insights
- Insights = Detections
- Detections = Protection

MINI LAB

- 1 Go to Logs.
- 2 Select table: SecurityEvent.
- 3 Set time range: Last 24 hours.
- 4 Run the query: `SecurityEvent | take 50`
- 5 Review the results and identify a few different EventIDs.

WHAT'S NEXT?

Now that you know what logs are, let's learn the basics of KQL so you can query like a pro!



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12. KQL BASICS

"KQL is Google for security logs."



Microsoft
Sentinel

KQL (Kusto Query Language) is used in Microsoft Sentinel to search, filter and analyze log data from all your connected sources.

EXAMPLE 1: VIEW LATEST SECURITY EVENTS

Microsoft Sentinel | Logs
Selected workspace: law-sentinel-lab

Tables Queries Functions x

Search
Filter
Group by: Solution Filters: not selected

Favorites
You can add favorites by clicking on the icon

Log management
All Logs
AzureActivity
CommonSecurityLog
Heartbeat
SecurityAlert
SecurityEvent
SigninLogs
Syslog
WindowsEvent
OfficeActivity
Custom Logs

1 1 SecurityEvent
2 | take 10
3

Results Chart

Completed. Showing results from the last 24 hours. 00:00:00.512 10 records

TimeGenerated [UTC]	Computer	EventID	LevelDisplayName	Account	IpAddress
5/14/2024, 10:24:15.123 AM	DC01	4624	Information	CONTOSO\jdoe	10.0.0.15
5/14/2024, 10:24:16.456 AM	DC01	4625	Information	CONTOSO\asmith	10.0.0.25
5/14/2024, 10:24:17.789 AM	DC01	4634	Information	CONTOSO\jdoe	10.0.0.15
5/14/2024, 10:24:18.912 AM	DC01	4720	Information	CONTOSO\svc_sql	10.0.0.30
5/14/2024, 10:24:20.111 AM	DC01	4672	Information	CONTOSO\jdoe	10.0.0.15

BREAKDOWN

- 1 SecurityEvent Table
Contains Windows security events like logons, logoffs, process creation, etc.
- 2 | (Pipe)
Passes the data from the left side to the next command.
- 3 take 10
Returns the first 10 records from the results.

★ Simple, powerful and fast!

EXAMPLE 2: VIEW SIGN-IN LOGS

- 1 SigninLogs → Table: SigninLogs (Azure AD sign-in logs)
- 2 | take 10 → Command: take 10 (show 10 records)

TimeGenerated [UTC]	UserPrincipalName	AppDisplayName	Status	IpAddress	Location
5/14/2024, 10:15:12 AM	jdoe@contoso.com	Microsoft 365 Portal	Success	20.40.60.10	Seattle, US
5/14/2024, 10:17:45 AM	asmith@contoso.com	Outlook	Success	20.40.60.12	Seattle, US
5/14/2024, 10:18:20 AM	jdoe@contoso.com	Microsoft Teams	Success	20.40.60.10	Seattle, US
5/14/2024, 10:19:33 AM	svc_backup@contoso.com	Azure Portal	Failure	103.25.15.8	Mumbai, IN

WHAT IS KQL?

- ✓ Query language used in Microsoft Sentinel (built on Azure Data Explorer).
- ✓ Fast, powerful and purpose-built for log analytics.
- ✓ Similar to SQL, but optimized for large-scale log data.

COMMON KQL KEYWORDS (YOU WILL USE OFTEN)

KEYWORD	WHAT IT DOES	EXAMPLE
where	Filters data based on a condition	SecurityEvent where EventID == 4624
summarize	Aggregates data (count, min, max, etc.)	SecurityEvent summarize count() by EventID
sort	Sorts data (asc or desc)	SecurityEvent sort by TimeGenerated desc
project	Selects specific columns	SecurityEvent project TimeGenerated, Account, EventID
take	Returns top N records	SecurityEvent take 10
distinct	Returns unique values	SecurityEvent distinct Account
extend	Creates new calculated columns	SecurityEvent extend LogonType = tostring(EventData)

QUICK KQL EXAMPLES

Count events by EventID
SecurityEvent
| summarize count() by EventID

Top 5 failed logons
SecurityEvent
| where EventID == 4625
| take 5

Unique users who logged in
SecurityEvent
| where EventID == 4624
| distinct Account

SOC ANALYST TIP

Start simple. Master the basics first. Complex queries come with practice!

REMEMBER

- KQL is case-insensitive.
- Use | to chain commands.
- Every query starts with a table.

MINI LAB

- 1 Go to Logs in Microsoft Sentinel.
- 2 Select SecurityEvent table.
- 3 Run: SecurityEvent | take 10
- 4 Change to SigninLogs | take 10
- 5 Explore the results.

NEXT STEP

Now that you know the basics, let's build real KQL queries for SOC investigations.



Want more detailed explanation, labs, real-world examples and guided mentorship for this topic?

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13. KQL FOR SOC ANALYSTS



Microsoft
Sentinel

"Good analysts search. Great analysts query."

KQL helps you quickly find signals in the noise. Here are some everyday queries every SOC analyst should know.

1 FAILED LOGINS (LAST 24 HOURS)

SigninLogs

```
| where TimeGenerated > ago(24h)
| where ResultType != 0
| project TimeGenerated, UserPrincipalName,
  IPAddress, Location, ResultType, ResultDescription
| order by TimeGenerated desc
| take 10
```

▶ Run Time range : Last 24 hours

TimeGenerated [UTC]	UserPrincipalName	IPAddresses	ResultType	ResultDescription
5/14/2024, 10:45:12 AM	asmith@contoso.com	203.0.113.45	50126	Invalid password
5/14/2024, 10:44:51 AM	jdoe@contoso.com	198.51.100.23	50126	Invalid password
5/14/2024, 10:43:21 AM	mwhite@contoso.com	203.0.113.45	50057	Account disabled
5/14/2024, 10:42:11 AM	jdoe@contoso.com	198.51.100.23	50126	Invalid password

→ Finds failed login attempts and helps detect brute force attacks.

KQL QUICK TIPS

- ✓ KQL is not case sensitive.
- ✓ Use ago(24h), ago(7d) for time filtering.
- ✓ Always start simple, then build complex.
- ✓ Use project to show only what you need.
- ✓ Use take 10, 20 while exploring results.



2 SUCCESSFUL LOGINS (LAST 24 HOURS)

SigninLogs

```
| where TimeGenerated > ago(24h)
| where ResultType == 0
| project TimeGenerated, UserPrincipalName,
  IPAddress, Location, DeviceDetail.browser
| order by TimeGenerated desc
| take 10
```

▶ Run Time range : Last 24 hours

TimeGenerated [UTC]	UserPrincipalName	IPAddresses	Location	DeviceDetail.browser
5/14/2024, 10:50:33 AM	asmith@contoso.com	20.40.60.10	Seattle, US	Chrome
5/14/2024, 10:48:21 AM	jdoe@contoso.com	20.40.60.10	Seattle, US	Edge
5/14/2024, 10:47:02 AM	mwhite@contoso.com	20.40.60.11	Redmond, US	Chrome
5/14/2024, 10:46:33 AM	bcarter@contoso.com	20.40.60.12	New York, US	Edge

→ Use to track user activity and validate normal behavior.

3 SEARCH BY USER

let user = "asmith@contoso.com";

SigninLogs

```
| where UserPrincipalName == user
| project TimeGenerated, ResultType, IPAddress,
  Location, DeviceDetail.operatingSystem
| order by TimeGenerated desc
| take 20
```

▶ Run Time range : Last 7 days

TimeGenerated [UTC]	ResultType	IPAddress	Location	DeviceDetail.operatingSystem
5/14/2024, 10:50:33 AM	0	20.40.60.10	Seattle, US	Windows 11
5/14/2024, 9:15:21 AM	0	20.40.60.10	Seattle, US	Windows 11
5/14/2024, 8:02:11 AM	50126	203.0.113.45	Seattle, US	Windows 10
5/14/2024, 7:50:09 AM	0	20.40.60.11	Redmond, US	Windows 11

→ Quickly investigate a specific user's sign-in activity.

COMMON OPERATORS

- == Equal to
- != Not equal to
- > Greater than
- < Less than
- >= Greater or equal
- <= Less or equal

4 SEARCH BY IP ADDRESS

let ip = "203.0.113.45";

SigninLogs

```
| where IPAddress == ip
| project TimeGenerated, UserPrincipalName,
  ResultType, Location, AppDisplayName
| order by TimeGenerated desc
| take 20
```

▶ Run Time range : Last 7 days

TimeGenerated [UTC]	UserPrincipalName	ResultType	Location	AppDisplayName
5/14/2024, 10:45:12 AM	asmith@contoso.com	50126	Seattle, US	Microsoft 365 Portal
5/14/2024, 10:43:21 AM	mwhite@contoso.com	50057	Seattle, US	Microsoft 365 Portal
5/14/2024, 10:12:11 AM	jdoe@contoso.com	0	Seattle, US	Outlook
5/14/2024, 9:58:19 AM	bcarter@contoso.com	0	Seattle, US	Teams

→ Track activity coming from a suspicious or unknown IP.

USEFUL TIPS

- ★ Save your queries.
- ★ Build query library.
- ★ Reuse and adapt.
- ★ Validate results.
- ★ Document findings.



— MORE HANDY QUERIES —

5 MULTIPLE FAILED LOGINS FROM SAME IP

SigninLogs

```
| where TimeGenerated > ago(1d)
| where ResultType != 0
| summarize FailedCount = count() by IPAddress
| where FailedCount > 10
| order by FailedCount desc
```

6 IMPOSSIBLE TRAVEL (SAME USER)

let timeWindow = 1h;

SigninLogs

```
| where TimeGenerated > ago(1d)
| where ResultType == 0
| summarize arg_max(TimeGenerated, *)
  by UserPrincipalName, Location
| sort by UserPrincipalName asc, TimeGenerated asc
```

7 EXTERNAL LOGINS

SigninLogs

```
| where TimeGenerated > ago(1d)
| where IdentityProvider == "LiveId"
| project TimeGenerated, UserPrincipalName,
  IPAddress, Location, AppDisplayName
| order by TimeGenerated desc
```

SOC ANALYST TIP



Practice these queries daily.
The more you query, the faster you'll find threats!

REMEMBER



- Data in = Insights out
- KQL is powerful
- Practice makes perfect

MINI LAB



- 1 Go to Logs in Sentinel.
- 2 Use SecurityEvent and SigninLogs.
- 3 Run each query above.
- 4 Modify time range to 7 days.
- 5 Save 2 queries to your favorites.

NEXT UP



Now that you can query like a pro, let's learn about Analytics Rules and how Sentinel detects threats automatically.



Want more detailed explanation, labs, real-world examples and guided mentorship for this topic?

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14. ANALYTICS RULES

"Rules are Sentinel's eyes."

Analytics rules continuously analyze your data and generate alerts when suspicious activity is detected.



Microsoft
Sentinel

Microsoft Azure | Search resources, services, and docs (G+)

admin@contoso.com

Home > Microsoft Sentinel

Microsoft Sentinel | Analytics

Selected workspace: law-sentinel-lab

General

- Overview (Preview)
- Logs
- News & guides
- Threat management
 - Incidents
 - Workbooks
 - Hunting
 - Notebooks
 - Entity behavior
 - Threat intelligence
- Content management
 - Content hub
 - Repositories
 - Community
- Configuration
 - Data connectors
 - Analytics
 - Watchlist
 - Automation
 - Settings

188 Active rules | 15 Enabled | 2 Disabled | 34 Templates | Guides & Feedback

Active rules | Rule templates | Anomaly rules (Preview)

+ Create | Refresh | Enable | Disable | Delete | More

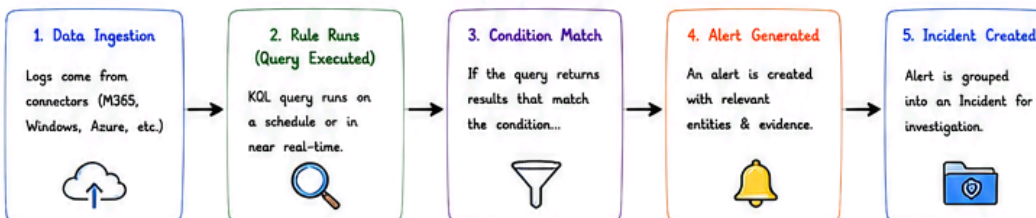
Search by ID, name, or description | Status: All | Rule type: All | Tactics: All

Rule name	Rule type	Status	Tactics	Last modified
☐ Multiple failed logons	Scheduled	Enabled	Credential Access	5/14/2024
☐ Impossible travel	Scheduled	Enabled	Initial Access	5/13/2024
☐ Rare sign-in location	Scheduled	Enabled	Initial Access	5/13/2024
☐ High volume email sends	Scheduled	Disabled	Exfiltration	5/10/2024
☐ Malware detected on host	Scheduled	Enabled	Execution	5/9/2024
☐ Admin role assigned	NRT	Enabled	Privilege Escalation	5/9/2024

TYPES OF ANALYTICS RULES

- Scheduled Rule**
Runs on a schedule (e.g., every 5 minutes, 1 hour). Best for bulk/periodic analysis.
Example: Multiple failed logins
- Near Real-Time (NRT) Rule**
Runs almost instantly on ingested data (1-2 min delay). Best for high priority detections.
Example: Admin role assigned
- Fusion Rule (Preview)**
Correlates alerts from multiple analytic rules or across workspaces. Best for advanced correlation.
Example: Brute force followed by impossible travel

HOW ANALYTICS RULES WORK



COMPARISON OF RULES TYPES

Rule Type	Execution	Frequency	Latency	Best For	Example
Scheduled	Runs on a schedule	Every 5m, 10m, 1h, etc.	5-15 minutes	Trends, patterns, aggregations	Multiple failed logins
Near Real-Time (NRT)	Runs on data ingestion	Continuous	~1-2 minutes	High priority threats, time-sensitive alerts	Admin role assigned
Fusion (Preview)	Correlates alerts/rules	Continuous	~1-5 minutes	Complex multi-stage attacks	Brute force + impossible travel

SOC ANALYST TIP

- ✓ Start with Scheduled rules while learning.
- ✓ Use NRT for high value alerts.
- ✓ Keep rules focused and specific.
- ✓ Use Fusion rules for advanced scenarios.

REMEMBER

- Analytics rules = Detections
- Good rules reduce noise and catch real threats.
- Test and tune your rules regularly.
- Context + Entities make investigations easier.

MINI LAB

- Go to **Analytics** in Microsoft Sentinel.
- Click **+ Create > Scheduled rule**.
- Select template: **"Multiple failed logons"**.
- Review the KQL query and conditions.
- Set frequency to **every 5 minutes**.
- Save and enable the rule.
- Wait for data and check for alerts!

WHAT TO LOOK FOR

- Spikes in failed logins
- Unusual sign-in locations
- Privilege changes
- Malware detections
- Excessive email sending

NEXT UP

Now that you understand Analytics Rules, let's create your first detection rule on the next page!



Want more detailed explanation, labs, real-world examples and guided mentorship for this topic?

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15. CREATE YOUR FIRST DETECTION RULE

"Teach Sentinel what suspicious looks like."



Microsoft
Sentinel

In this lab, you will create a rule that detects multiple failed logon attempts from the same user within 5 minutes.

OBJECTIVE: Create a Scheduled rule to detect 5 or more failed logons by the same user in 5 minutes.

Microsoft Sentinel Search resources, services, and docs (Gv7) admin@contoso.com CONTOSO

Home > Microsoft Sentinel > Analytics

Create analytic rule - Scheduled

1 General 2 Set rule logic 3 Incident settings 4 Automated response 5 Review + create

General

1 Rule name * Multiple failed logon attempts by user

2 Description Detects 5 or more failed logon attempts by the same user within 5 minutes.

Tactics Credential Access

3 Severity * High

Status Enabled

MITRE ATT&CK®

Techniques T1110 - Brute Force

Rule query View query results >

4 Query scheduling

Run query every 5 Minutes

Lookup data from the last 5 Minutes

5

Rule query

Query help

```
1 SecurityEvent
2 | where EventID == 4625
3 | summarize FailedCount = count() by
4   UserPrincipalName, IPAddress,
5   bin(TimeGenerated, 5m)
6 | where FailedCount >= 5
7 | project TimeGenerated, UserPrincipalName,
8   IPAddress, FailedCount
```

View results Chant

Completed. Showing results from the last 24 hours. 00:00:01.234 5 records

TimeGenerated [UTC]	UserPrincipalName	IPAddress	FailedCount
5/14/2024, 10:25:00 AM	jdoe@contoso.com	203.0.113.45	8
5/14/2024, 10:20:00 AM	jdoe@contoso.com	203.0.113.45	7
5/14/2024, 10:15:00 AM	asmith@contoso.com	198.51.100.23	6
5/14/2024, 10:10:00 AM	mwhite@contoso.com	203.0.113.45	5
5/14/2024, 10:05:00 AM	jdoe@contoso.com	203.0.113.99	5

Next: Set rule logic > Cancel

STEP-BY-STEP

- 1 Name your rule**
Give it a clear and descriptive name.
- 2 Add a description**
Explain what the rule detects.
- 3 Set severity**
Choose High for failed logon brute force.
- 4 Select tactic**
Credential Access is the right fit.
- 5 Set scheduling**
 - Run query every: 5 min
 - Lookup data from last: 5 min
- 6 Write the KQL query**
This query looks for 5 or more failed logons by the same user.
- 7 Click "Next"**
Proceed to Incident settings to create incidents when the rule triggers.



KQL EXPLAINED

```
1 SecurityEvent
2 | where EventID == 4625
3 | summarize FailedCount = count() by
4   UserPrincipalName, IPAddress,
5   bin(TimeGenerated, 5m)
6 | where FailedCount >= 5
7 | project TimeGenerated, UserPrincipalName,
8   IPAddress, FailedCount
```

LINE	WHAT IT DOES	WHY IT MATTERS
1	Uses SecurityEvent log (Windows security logs)	Source of logon events
2	Filters only failed logon events (Event ID 4625)	We only care about failures
3-5	Groups events by user, IP and 5-minute time window	Helps count attempts in short bursts
6	Keeps only groups with 5 or more failures	Detects brute force behavior
7-8	Shows the time, user, IP and number of failures	This becomes the alert evidence

SOC ANALYST TIP



- ✓ Start simple, then make rules more advanced.
- ✓ Test your query with real data before saving.
- ✓ Think like an attacker.
- ✓ 5 failed logons in 5 minutes is suspicious!

WHAT HAPPENS WHEN THIS RULE TRIGGERS?



1. Rule Runs

Query runs every 5 minutes.



2. Condition Met

5+ failed logons detected.



3. Alert Generated

Alert is created with evidence.



4. Incident Created

Alert is grouped into an Incident.



5. Analyst Investigates

You investigate, respond and close the incident.

MINI LAB



- Go to Analytics in Sentinel.
- Click "+ Create" > "Scheduled rule".
- Name your rule.
- Use the query provided.
- Set schedule to every 5 minutes.
- Click "Next" and complete Incident settings.
- Create the rule and test!

TRY IT OUT



Change the query to detect 10 or more failed logons in 10 minutes.

What happens to the results?

COMMON MISTAKES



- ✗ Not filtering correct EventID
- ✗ Using too short time windows
- ✗ Setting rule severity too low
- ✗ Not testing the query
- ✗ Not reviewing incidents

NEXT UP



Now that the rule is ready, let's see how incidents are created and investigated in the next page!



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16. INCIDENTS EXPLAINED



Microsoft
Sentinel

"An incident is Sentinel asking for your attention."

Incidents are alerts that have been grouped together by Sentinel and require investigation. They give you the full story, not just a single alert.

MICROSOFT SENTINEL → INCIDENTS

Incidents

121 Open incidents | 35 New incidents | 23 Active incidents | 63 Closed incidents | Last 24 hours

+ Create incident | Refresh | Last 24 hours | Actions | Security efficiency workbook

Search by ID, name, owner or product | Severity: All | Status: All | Owner: All | Product name: All

Incident name	Severity	Status	Owner	Created time
Multiple Failed Login Attempts	High	New	Unassigned	5/14/2024, 10:25 AM
Impossible Travel - User Login	Medium	Active	john.doe	5/14/2024, 09:48 AM
Suspicious PowerShell Activity	High	Active	sarah.analyst	5/14/2024, 09:15 AM
Malware Detected on Host	High	Closed	mike.soc	5/14/2024, 08:32 AM
Unusual Sign-in Location	Medium	Closed	sarah.analyst	5/14/2024, 07:55 AM

WHAT IS AN INCIDENT?

An incident is a group of related alerts that represent a potential threat or attack.

WHAT IS AN ALERT?

An alert is a signal triggered by a rule when something suspicious happens.

ALERT vs INCIDENT

- Alert = Signal
- Incident = Story (Multiple alerts + context).

1 INCIDENT NAME

The name of the incident. It tells you what happened.

2 SEVERITY

How serious the incident is based on impact. High, Medium or Low.

3 STATUS

Where the incident is in its lifecycle. New, Active or Closed.

4 OWNER

Who is responsible for investigating this incident.

5 CREATED TIME

When the incident was created by Sentinel.

EXAMPLE INCIDENT



Name : Multiple Failed Login Attempts

Severity : High

Status : New

Owner : Unassigned

Created : 5/14/2024, 10:25 AM

WHAT HAPPENED?

- The same user tried to login many times and failed.
- This can be a sign of password guessing or brute force attack.
- Needs immediate attention!



WHY INCIDENTS MATTER

- ✓ They reduce noise.
- ✓ Give full context.
- ✓ Help you prioritize work.
- ✓ Improve response time.
- ✓ Better visibility for SOC.



MINI LAB



- Go to Incidents in Microsoft Sentinel.
- Look at the list and identify one High severity incident.
- Identify one Medium and one Low severity incident.
- Check their Status and Created Time.
- Open one incident and explore.

→ Practice helps you become faster!

INCIDENT SEVERITY GUIDE

- High - Critical impact. Immediate action needed.
- Medium - Moderate impact. Investigate soon.
- Low - Low impact. Monitor and review.
- Informational - Very low impact. For awareness.



Start with High severity incidents first!

ANALYST TIP



Don't just open incidents. Ask these questions first:

- What is the impact?
- What systems are affected?
- Who is the attacker?
- What should I do next?



Incidents are the heart of SOC work. Learn to read, understand and act quickly!

Visit blueteamers.io for labs, mentorship and SOC training.





17. INVESTIGATING INCIDENTS



Microsoft
Sentinel

"Every incident tells a story."

When an incident is created, your job is to investigate, understand what happened, find the attacker's path and stop the attack.

INCIDENT INVESTIGATION SCREEN (OVERVIEW)

The screenshot shows the Microsoft Sentinel interface. The top navigation bar includes 'Home', 'Microsoft Sentinel', 'Incidents', and 'Incident'. The main content area displays details for an incident titled 'Multiple Failed Login Attempts' with ID 12345. The incident is classified as 'High' severity, 'New' status, 'Unassigned' owner, and was created on '5/14/2024, 10:25 AM'. The left sidebar shows various management options like Overview, Logs, Threat management, and Configuration. The main panel is divided into sections: Description (user 'john.doe@contoso.com' performed failed login attempts from IP 203.0.113.45), Key info (Incident ID, Product, Primary entity, First/Last activity, Total alerts/entities), and a 'Classify incident' section with dropdowns for Classification, Category, Subcategory, and Impact.

WHAT YOU WILL DO

- ✓ Understand what happened.
- ✓ Find who is involved.
- ✓ See attacker's actions (timeline).
- ✓ Collect evidence.
- ✓ Decide the impact.
- ✓ Take action & close the incident.



INCIDENT INVESTIGATION COMPONENTS

1 OVERVIEW



Quick summary of the incident.

- Description
- Key information
- Classification

2 TIMELINE



Shows the order of events.

- When things happened
- What triggered the alert
- How the attacker moved

3 ENTITIES



Who/What is involved.

- Users
- Hosts
- IP Addresses
- Apps, Devices

4 ALERTS



All alerts that are part of this incident.

- Which rule triggered
- When
- Severity

5 EVIDENCE



Artifacts related to the attack.

- Logs
- Files
- Commands
- URLs

6 RELATED EVENTS



Other events that may be connected.

- Helps you understand the full attack path.

SOC ANALYST WORKFLOW

1

OPEN

Open the incident and read the summary.

2

ANALYZE

Check timeline, alerts, entities and evidence.

3

VERIFY

Validate if it is a real threat or false positive.

4

RESPOND

Take action: block IP, reset password, isolate host, etc.

5

CLOSE

Document findings, close incident and learn from it.

INCIDENT STORY (EXAMPLE)



Attacker tries password spraying from IP 203.0.113.45



Analytics rule detects multiple failed logins



Alert triggers



Incident is created



You investigate, collect evidence and understand the attack



You stop the attack and close the incident

MINI LAB



- Go to Microsoft Sentinel.
- Click on "Incidents".
- Open an incident.
- Identify incidents with:

- High severity
- Medium severity
- Low severity

- Read the overview.

- Check timeline, entities and alerts.

- Write down what happened.

- Goal: Understand how incidents look and how to open them.

ANALYST TIP



Always read the Overview first. It gives you 80% of the context in 20% of the time.



Investigation is about asking the right questions and following the evidence. Practice every day. You will get faster!





18. ENTITY BEHAVIOR



Microsoft
Sentinel

"Who did what?"

Entities represent the who, what, where and how in your environment. Understanding entity behavior helps you investigate faster and deeper.

ENTITY PAGE IN MICROSOFT SENTINEL (EXAMPLE: USER ENTITY)

The screenshot shows the Microsoft Sentinel interface for a User entity. The left sidebar contains navigation options like Home, Microsoft Sentinel, Incidents, Multiple Failed Login Attempts, and Entities. The main content area displays the entity's overview, including first and last seen times, account type, and risk level. It also features sections for related alerts, incidents, and logon activity, along with top information such as UPN, display name, department, location, risk score, last logon IP, last logon location, and authentication method.

WHAT IS AN ENTITY?

An entity is any identifiable object in your environment.

Examples:

- User
- Host
- IP Address
- Application
- File

WHY IT MATTERS

- ✓ Know who is involved.
- ✓ See what they did.
- ✓ Find related alerts and incidents.
- ✓ Understand the impact.
- ✓ Speed up investigations.



TYPES OF ENTITIES

USER ENTITY

Represents a user in Azure AD or on-prem. Includes logons, roles, devices, locations, etc.

Example: john.doe@contoso.com

HOST ENTITY

Represents a machine (server, workstation, VM, etc.). Includes processes, events, vulnerabilities.

Example: WS-01.contoso.com

IP ENTITY

Represents an IP address (source or destination). Includes geolocation, connections, threats.

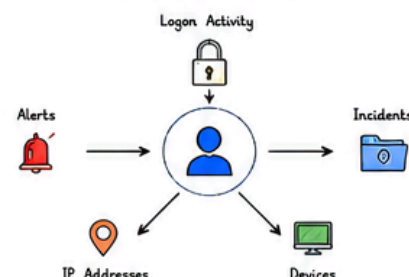
Example: 203.0.113.45

APP ENTITY

Represents an application or service. Includes sign-ins, usage, permissions.

Example: 203.0.113.45

EXAMPLE: USER JOHN.DOE@CONTOSO.COM



USER ENTITY - ACTIVITY OVERVIEW

LOGIN ACTIVITY

- Successful logins
- Failed logins
- MFA usage
- Locations & IPs

ALERTS

- Failed login attempts
- Impossible travel
- Brute force detected
- Suspicious sign-ins

INCIDENTS

- Related incidents
- Incident roles
- Investigation status
- Impact

IP ADDRESSES

- Recent IPs used
- Common locations
- New / Unusual IPs
- Threat intel matches

DEVICES

- Devices used
- OS / Browser info
- Compliance status
- Risk score

REAL WORLD EXAMPLE

An attacker compromises john.doe@contoso.com and tries to login from an unusual location.

By checking the entity, you can see:

- New IP
- Failed logins
- Related alerts
- Impacted devices



MINI LAB



- Go to Microsoft Sentinel.
- Click on "Entity behavior".
- Choose "Users".
- Search for a user (e.g., john.doe@contoso.com).
- Review the overview, alerts and incidents.
- Check logon activity and IP addresses.

→ Understand the story behind the user!

ANALYST TIP



- ✓ Always start investigation from the entity.
- ✓ Entities give full context and history.
- ✓ Check first seen, last seen and risk level.
- ✓ Look at related alerts and incidents.
- ✓ This saves hours of investigation time.

REMEMBER



- Entities = Context
- Context = Clarity
- Clarity = Faster Response
- Faster Response = Stronger SOC!



Know your entities. Know your environment. Know your enemy. Practice every day and become a better analyst!





19. WORKBOOKS & DASHBOARDS

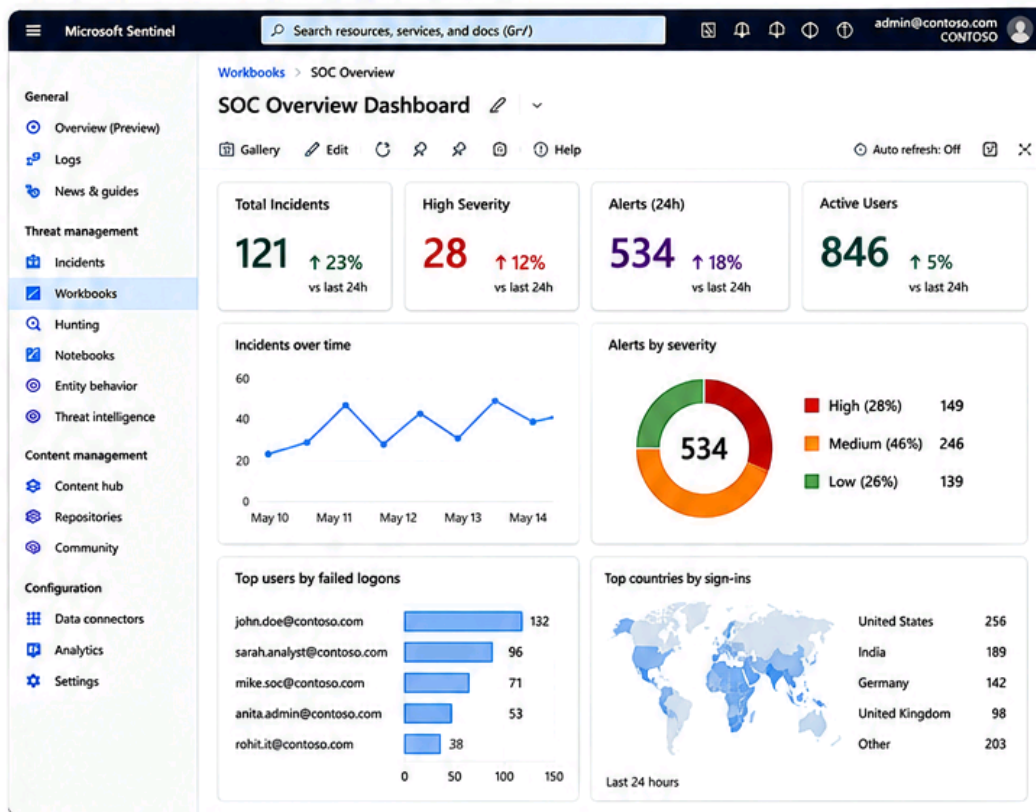


Microsoft
Sentinel

"Turn logs into visibility."

Workbooks help you visualize data, track security trends and build beautiful dashboards for SOC monitoring and reporting.

MICROSOFT SENTINEL WORKBOOK (EXAMPLE)



WHAT IS A WORKBOOK?

A workbook is a custom dashboard built with queries, visualizations and text to monitor and analyze data.

WHAT CAN YOU ADD?

- Charts
- Tables
- KPIs
- Filters
- Text & Notes
- Images
- Maps

WHY USE WORKBOOKS?

- ✓ Better visibility
- ✓ Faster analysis
- ✓ Easy reporting
- ✓ Real-time monitoring
- ✓ Share with team

WORKBOOK COMPONENTS

- KPI**: Shows key numbers at a glance.
- CHART**: Visualize trends and patterns.
- TABLE**: Show detailed data in rows & columns.
- MAP**: See activity across countries/regions.
- TEXT**: Add notes, context and explanations.
- FILTER**: Allow users to filter data (time, user, etc.).

EXAMPLE WORKBOOKS FOR SOC

FAILED LOGINS

Track failed login attempts

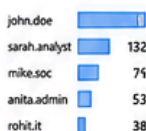


3,214

Last 24 hours

TOP USERS

Who is generating most alerts?



TOP COUNTRIES

Where are sign-ins coming from?



United States	256
India	189
Germany	142
United Kingdom	98
Other	203

INCIDENT TRENDS

Incidents over last 7 days



121

Total Incidents

ALERT TRENDS

Alerts over last 7 days



534

Total Alerts

MINI LAB



- Go to Microsoft Sentinel.
- Click on "Workbooks".
- Click "+ New workbook".
- Add a query (use example below).
- Add a chart to visualize results.
- Save the workbook.
- Pin it to your workspace.

→ Share with your team!

SAMPLE KQL FOR WORKBOOK

// Failed Logons - Last 24 Hours

```
SecurityEvent
| where EventID == 4625
| summarize FailedCount = count()
  by bin(TimeGenerated, 1h)
| render timechart
```

Use this query in your workbook to create a chart.



WORKBOOK TIPS



- ✓ Keep it simple and focused.
- ✓ Use clear titles and descriptions.
- ✓ Add filters for time, users, etc.
- ✓ Use consistent colors.
- ✓ Review and update regularly.
- ✓ Make it actionable!



Collect Logs
Via connectors



Query Data
Using KQL



Visualize
In Workbook



Understand
What's happening



Take Action
Stop threats!

REMEMBER



- Dashboards don't stop attacks. People do.
- Good visibility makes you faster!



Great dashboards don't just look good - they help you see what matters!

Build. Visualize. Investigate. **Respond.**





20. AUTOMATION RULES



Microsoft
Sentinel

"Stop doing repetitive work."

Automation rules help you automatically perform actions when specific conditions are met. They save time, reduce manual work and ensure consistent response.

AUTOMATION RULES IN MICROSOFT SENTINEL

Microsoft Sentinel

Search resources, services, and docs (Gr/)

admin@contoso.com
CONTOSO

Home > Microsoft Sentinel > Automation

Automation

Create automation rules to take actions on alerts and incidents

+ Create Refresh Enable Disable Delete Columns

1 Rule name	2 Trigger	3 Status	4 Actions	5 Last modified
1 Assign High Severity Incidents	When incident is created Severity = High	Enabled	Assign to group (SOC) Add tag: HighPriority	5/14/2024, 10:25 AM
2 Send Email for Critical Alerts	When alert is triggered Severity = High	Enabled	Send email to SOC team	5/14/2024, 09:50 AM
2 Create Ticket in ServiceNow	When incident is created Severity = High or Medium	Enabled	Create ServiceNow incident	5/13/2024, 04:15 PM
4 Close Low Severity Incidents	When incident is closed Severity = Low	Enabled	Add tag: Closed-Low Send email (summary)	5/13/2024, 03:20 PM
2 Block Malicious IP (Playbook)	When alert is triggered IP is malicious	Enabled	Run playbook (Block IP)	5/12/2024, 11:40 AM

WHAT IS AN AUTOMATION RULE?

A rule that triggers actions automatically when a condition is met.

BENEFITS

- ✓ Save time
- ✓ Reduce manual work
- ✓ Consistent response
- ✓ Faster reaction
- ✓ Less human error



1 RULE NAME

The name that describes what the rule does.

2 TRIGGER

The condition that starts the rule (Incident/Alert/Event).

3 STATUS

Shows whether the rule is Enabled or Disabled.

4 ACTIONS

The tasks Sentinel will perform automatically.

5 LAST MODIFIED

The date and time when the rule was last updated.

EXAMPLE: HIGH SEVERITY INCIDENT → AUTO RESPONSE



Trigger

Incident created
Severity = High



Condition

If Severity is High



- Assign to SOC group
- Add tag: HighPriority
- Send email to SOC team
- Create ticket in ServiceNow



Result

Analyst notified,
ticket created,
investigation starts!

SOC USE CASE



High severity incident created
↓
Rule triggers automatically
↓
Assign to analyst
↓
Send email notification
↓
Create ticket in ServiceNow
↓
Investigation starts **faster!**

HOW TO CREATE AN AUTOMATION RULE



- Go to Microsoft Sentinel.
- Click on "Automation".
- Click "+ Create".
- Give a name and description.
- Select the trigger (incident/alert).
- Add conditions (severity, status, etc.).
- Add actions (email, assign, playbook, ticket).
- Review and click "Create".



REAL WORLD EXAMPLE

Scenario

Multiple failed login attempts from a suspicious IP.

Automation Rule

When alert is triggered
Severity = High
IP is suspicious

Actions

- Run playbook (Block IP)
- Send email to SOC
- Add tag to incident
- Create ticket

Outcome

Attacker blocked automatically and analysts are notified instantly!

ANALYST TIP



Start small. Automate the repetitive tasks first. This will give you the biggest impact in SOC!



MINI LAB



- Go to Automation in Sentinel.
- Create a rule: When incident is created and Severity = High.
- Action: Send email to your team.
- Test the rule by generating a high severity alert.
- Verify that email is received!

BEST PRACTICES

- ✓ Automate repetitive and low-value tasks.
- ✓ Always test your rule before enabling.
- ✓ Use clear names and descriptions.
- ✓ Monitor rule execution and failures.
- ✓ Keep rules updated as environment changes.

REMEMBER



- Automate Smartly
- Respond Quickly
- Save Time
- Protect Better



Automation is not about replacing analysts, it's about empowering them!

Use automation → Focus on what matters → Stop the attackers.





21. PLAYBOOKS & SOAR



Microsoft
Sentinel

"What if Sentinel could respond before
you even touch the keyboard?"

Playbooks (powered by Logic Apps) let Sentinel take automatic action when alerts or incidents are triggered. This is SOAR in action!

▶ WHAT IS SOAR?

SOAR (Security Orchestration, Automation and Response) automates repetitive tasks and speeds up response.

▶ WHAT IS A PLAYBOOK?

A playbook is a Logic App workflow that runs automated steps when a trigger occurs.

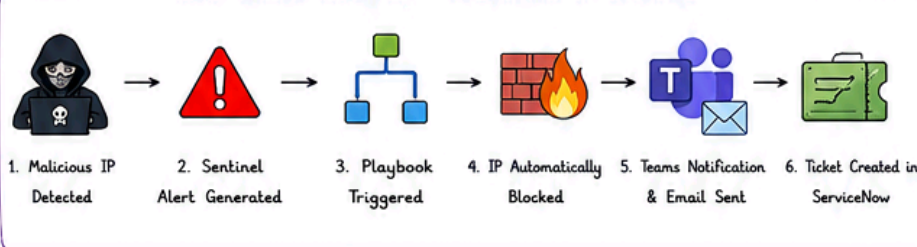
▶ WHY SOC TEAMS USE PLAYBOOKS?

- ✓ Auto respond in seconds
- ✓ Reduce manual work
- ✓ Consistent & accurate actions
- ✓ Integrate with tools
- ✓ Free analysts to focus on real threats

MICROSOFT SENTINEL – PLAYBOOKS (EXAMPLE)

1 TRIGGER	2 LOGIC APP (PLAYBOOK)	3 ACTION	4 RESPONSE	5 STATUS
When incident is created (High)	Block IP and Notify (SOC Playbook)	- Extract IP - Check reputation - Block IP	- Teams + Email - ServiceNow Ticket	Enabled
When alert is triggered (Malicious IP)	Malicious IP Response	- Get IP details - Block IP in Firewall - Add IOC	- Teams Notification - Email to SOC	Enabled
When incident is closed	Close Incident Tasks	- Update ticket - Add notes - Notify owner	- Email to Owner - Update in Teams	Enabled

REAL WORLD EXAMPLE – MALICIOUS IP DETECTED



EXPLAINING THE PLAYBOOKS SCREEN

1 TRIGGER	2 LOGIC APP (PLAYBOOK)	3 ACTION	4 RESPONSE	5 STATUS
Defines what starts the playbook. Examples: • Incident Created • Alert Triggered • Incident Closed • Entity Matched	The actual workflow built in Azure Logic Apps that contains the steps and logic.	Tasks performed by the playbook. Examples: • Get IP Details • Block IP • Add IOC • Update Ticket	Where the results are sent. Examples: • Microsoft Teams • Email • ServiceNow • Webhook	Shows whether the playbook is currently Enabled or Disabled.

HOW IT WORKS (STEP-BY-STEP)

- 1 Trigger**
Alert/Incident occurs based on analytics rule.
- 2 Playbook Starts**
Sentinel calls the Logic App (playbook).
- 3 Actions Execute**
Playbook performs API calls, checks, and automated tasks.
- 4 Response Sent**
Notifications, blocking, ticket creation, updates – all auto.
- 5 Analyst Notified**
Analyst gets full context and can take further action if needed.

MINI LAB

- Go to Microsoft Sentinel.
 - Click on "Automation" from the left menu.
 - Open "Playbooks".
 - Choose any playbook (e.g., Block IP and Notify).
 - Click "Run" (test with a sample alert/IP).
 - Check the Logic App run history.
 - Verify the action (IP blocked / Email / Ticket).
- You just ran your first Playbook!

SOAR BENEFITS

- ✓ Respond in seconds, not minutes.
- ✓ Eliminate repetitive manual tasks.
- ✓ Reduce risk of human error.
- ✓ Improve SOC efficiency.
- ✓ Scale operations without adding more people.

COMMON PLAYBOOK USE CASES

- Block malicious IP addresses
- Disable compromised user accounts
- Add indicators to threat intelligence
- Enrich alerts with external data
- Create / Update tickets automatically
- Notify analysts and stakeholders

ANALYST TIP

Start small! Automate simple tasks first (like IP blocking + notification). Then build advanced playbooks as you gain confidence.



Playbooks won't replace analysts → they **empower analysts!**
Automate the noise. Focus on the **real threats.**



22. THREAT HUNTING



Microsoft
Sentinel

"Great analysts don't wait for alerts."

Threat hunting is a proactive approach. You search through data to find threats that may have bypassed alerts. You think like an attacker and find the unknown.

▶ WHAT IS THREAT HUNTING?

Proactively searching through logs and data to find suspicious or malicious activity that has not triggered alerts.

▶ WHY IT MATTERS

- ✓ Attackers avoid detections
- ✓ Find hidden threats
- ✓ Reduce dwell time
- ✓ Stop attacks early
- ✓ Improve security posture

▶ ALERT INVESTIGATION VS THREAT HUNTING

ALERT INVESTIGATION	THREAT HUNTING
Reacts to alerts	Proactively searches
Known threats	Unknown threats
Start with something (Signal)	Start with hypothesis (Question)
Focused on one event	Looks across many events

MICROSOFT SENTINEL – HUNTING PAGE (EXAMPLE)

HUNTING TIPS

- ✓ Think like an attacker.
- ✓ Start with a hypothesis.
- ✓ Use KQL to search.
- ✓ Look for anomalies.
- ✓ Pivot and investigate.
- ✓ Document & share findings.



COMMON HUNTING SCENARIOS & KQL EXAMPLES

1 BRUTE FORCE SIGN-IN ATTEMPTS



Detect multiple failed sign-ins from the same IP.

```
SigninLogs
| where ResultType != 0
| summarize FailedCount = count()
  by IPAddress, bin(TimeGenerated, 5m)
| where FailedCount > 10
| order by FailedCount desc
```

What it finds:

IPs trying many passwords in a short time.

2 IMPOSSIBLE TRAVEL (GEO ANOMALY)



Detect sign-ins from two distant locations in a short time.

```
let timeWindow = 30m;
let locations = SigninLogs
| where ResultType == 0
| summarize make_set(Location, 10),
  count() by UserPrincipalName,
  bin(TimeGenerated, timeWindow)
| where count_ > 1 and array_length(
  make_set_Location) > 1
```

What it finds:

Same user logged in from different countries in a short time.

3 POWERSHELL ABUSE



Look for suspicious PowerShell commands and encoded scripts.

```
SecurityEvent
| where EventID == 4688
| where NewProcessName has "powershell.exe"
| where ProcessCommandLine has_any
  ("enc", "--EncodedCommand", "IEX ",
  "DownloadString", "Invoke-Expression")
| project TimeGenerated, Computer,
  Account, ProcessCommandLine
```

What it finds:

Encoded commands, downloads, IEX, suspicious PowerShell usage.

4 SUSPICIOUS ADMIN ACTIVITY



Detect abnormal admin actions and privilege changes.

```
AuditLogs
| where Category in ("RoleManagement",
  "Add member to role", "Role assignment")
| where TimeGenerated > ago(7d)
| project TimeGenerated, OperationName,
  TargetResources, InitiatedBy
```

What it finds:

Role changes, new admin assignments, privilege escalation.

THREAT HUNTING WORKFLOW



1. Hypothesize
Start with a question.



2. Search
Use KQL to query data.



3. Analyze
Look for patterns and anomalies.



4. Pivot
Dig deeper, follow the trail.



5. Validate
Confirm if it's malicious.



6. Report
Document and share findings.

SAMPLE HUNTING MINDSET



- What would I do if I were an attacker?
- What data would I touch?
- How would I persist?
- What would be unusual?

Ask questions. Use data. Find the truth.

MINI LAB



- Go to Microsoft Sentinel.
- Click on "Hunting".
- Select a query (e.g., Brute Force).
- Review the results.
- Modify the query and run again.
- Save the query to My Queries.

→ You just hunted like a pro!

HUNTING DATA SOURCES



Azure AD
(SigninLogs)



Windows Events
(SecurityEvent)



Office 365
(AuditLogs)



Firewall / Network
(Syslog)



Cloud Apps
(CloudAppEvents)

The more data, the better the hunt!

SENIOR ANALYST TIP



"Hunting is not about tools. It's about curiosity, creativity and consistency."

Practice every day!



Alerts tell you what happened. Hunting tells you what they didn't catch.
Be proactive. Stay curious. **Hunt threats!**





23. REAL INCIDENT CASE STUDY



Microsoft
Sentinel

IMPOSSIBLE TRAVEL ATTACK "India at 9 AM. Germany at 9:05 AM. Impossible."

Attackers use stolen credentials to login from two distant locations in a short time.

Sentinel detects the impossible travel and alerts you.

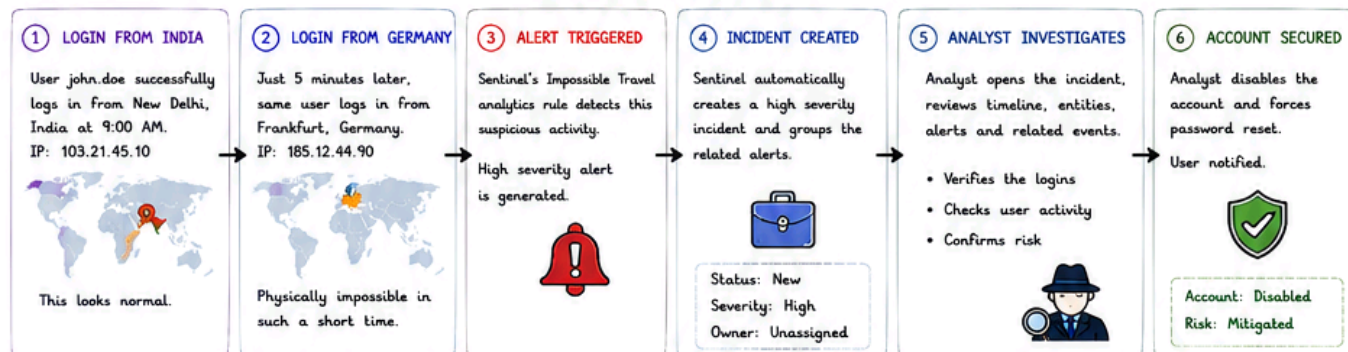
THE ATTACK STORY

- 1 User Login - India**
9:00 AM
User john.doe logs in from IP 103.21.45.10 (New Delhi, India)
- 2 5 Minutes Later**
9:05 AM
Same user logs in from IP 185.12.44.90 (Frankfurt, Germany)
- 3 Impossible Travel**
Impossible Travel Detected by Sentinel Analytics Rule
- 4 Incident Created**
High severity incident created automatically
- 5 Analyst Investigates**
SOC analyst investigates the incident
- 6 Account Secured**
Account disabled & user notified

MICROSOFT SENTINEL - INCIDENT DETAILS (EXAMPLE)

The screenshot shows the Microsoft Sentinel interface. The incident is titled "Impossible travel to multiple countries" with ID 7f3b7a2e-6bb6-4a8a-9f2e-3b1c97c5e6d0. It is a high severity incident created on 5/14/2024 at 9:07 AM. The incident timeline shows two logins: one from New Delhi, India at 9:00:12 AM and another from Frankfurt, Germany at 9:05:21 AM, separated by a 5-minute gap. An alert was triggered at 9:07:03 AM, and the incident was created at 9:07:05 AM. The console also shows 2 alerts, 1 entity (User), and the source as Azure AD SignInLogs.

EXPLAINING EVERY STEP



INCIDENT TIMELINE & EVIDENCE (INSIDE SENTINEL)

Overview	Alerts (2)	Entities (1)	Timeline	Evidence	Bookmarks	Related incidents
9:00:12 AM	User logged in		Success	New Delhi, India	103.21.45.10	
9:01:05 AM	User accessed Azure Portal		Success	New Delhi, India	103.21.45.10	
9:05:21 AM	User logged in		Success	Frankfurt, Germany	185.12.44.90	
9:05:45 AM	Risky sign-in detected		Medium	Frankfurt, Germany	185.12.44.90	
9:07:03 AM	Impossible travel detected		High	-	-	
9:07:05 AM	Incident created		High	-	-	

Entities (1): john.doe@contoso.com User. View full details >

Evidence: 2 Alerts, 18 Events. View all evidence >

WHY THIS MATTERS

- ✓ Attackers use stolen creds.
- ✓ Impossible travel is a strong indicator.
- ✓ Early detection stops account takeover.
- ✓ Fast investigation = less damage.

KQL BEHIND THE DETECTION

```
// Simplified KQL for impossible travel
SigninLogs
| where TimeGenerated > ago(1d)
| where ResultType == 0
| summarize arg_min(TimeGenerated, *)
  by UserPrincipalName, bin(TimeGenerated, 5m)
| extend Country = LocationDetails.countryOrRegion
| summarize FirstLocation = any(Location),
  FirstTime = min(TimeGenerated),
  LastLocation = any(Location),
  LastTime = max(TimeGenerated)
  by UserPrincipalName
| where FirstLocation != LastLocation
  and datetime_diff('minute', LastTime, FirstTime) < 60
```

ANALYST ACTIONS TAKEN

- ✓ Opened the incident.
- ✓ Reviewed alerts and timeline.
- ✓ Checked user activity and IP details.
- ✓ Confirmed impossible travel.
- ✓ Disabled the account.
- ✓ Forced password reset.
- ✓ Notified user and stakeholders.
- ✓ Documented and closed the incident.

LESSONS LEARNED

- Enable impossible travel detection.
- Monitor high risk users closely.
- Multi-factor authentication helps.
- Review alerts quickly.
- Investigate context, not just alerts.
- Respond fast and document everything.

Real attacks are fast. Good analysts are faster.

Detect. Investigate. Contain. Repeat.



24. MICROSOFT SENTINEL INTERVIEW QUESTIONS



Microsoft
Sentinel

"Can you answer these without Google?"

MOST COMMON INTERVIEW QUESTIONS & ANSWERS

1 What is Microsoft Sentinel?

- Microsoft Sentinel is a cloud-native SIEM and SOAR solution that collects data, detects threats, investigates incidents and responds automatically.

2 What is Log Analytics Workspace?

- It is a container that stores log data from various sources in Azure Monitor and is required to use Sentinel.

3 What is KQL?

- KQL (Kusto Query Language) is a powerful query language used to search, analyze and visualize data in Sentinel.

4 What is an Incident?

- An incident is a grouping of related alerts that represent a potential security threat.

5 What is an Analytics Rule?

- Analytics rules are custom detections written in KQL that run on scheduled intervals to identify suspicious activity.

6 What is an Automation Rule?

- Automation rules trigger actions (playbooks) when specific conditions are met on alerts or incidents.

7 What is a Playbook?

- A playbook is an automated workflow (Logic App) that performs actions such as blocking IP, sending email, creating tickets, etc.

8 What is SOAR?

- SOAR (Security Orchestration, Automation and Response) automates repetitive tasks and accelerates incident response.

9 What is Threat Hunting?

- Threat hunting is the proactive search for threats that may not have triggered any alerts.

10 What is the difference between Alert and Incident?

- Alert is a single detection. Incident is a collection of related alerts that represent a threat.

IMPORTANT DIFFERENCES

ALERT vs INCIDENT



ALERT

VS



INCIDENT

- Single detection
- Generated by rule
- Contains low context
- Example: 1 failed login
- Group of related alerts
- Represents a threat
- Contains full context
- Example: Brute force attack

EVENT vs ALERT



EVENT

VS



ALERT

- Raw log record
- Generated by systems
- Very low context
- Example: 4625 log
- Generated by rule
- Has context & meaning
- Indicates suspiciousness
- Example: Multiple failed logins from same IP

SIEM vs SOAR



SIEM

VS



SOAR

- Collects and analyzes data
- Detects & visualizes threats
- Example: Microsoft Sentinel
- Automates response
- Orchestrates actions
- Example: Playbooks, Automation Rules

INTERVIEW CHECKLIST

- ✓ Understand Microsoft Sentinel Architecture
- ✓ Know Log Analytics Workspace
- ✓ Write KQL Queries confidently
- ✓ Create Analytics & Automation Rules
- ✓ Investigate Incidents end-to-end
- ✓ Use Playbooks for response
- ✓ Perform Threat Hunting
- ✓ Build Workbooks & Dashboards
- ✓ Understand Connectors & Data Sources
- ✓ Know Azure / M365 / AD logs



KQL QUICK RECAP

```
// Top failed logons by IP
SecurityEvent
| where EventID == 4625
| summarize count() by
  IPAddress
| sort by count_desc
```

```
// Impossible travel
SignInLogs
| where TimeGenerated > ago(1d)
| summarize min(TimeGenerated),
  max(TimeGenerated) by UserPrincipalName
| extend diff_min = datetime_diff('minute',
  max_TimeGenerated, min_TimeGenerated)
| where diff_min < 60 and
  dcount(Location) > 1
```

TIPS TO ANSWER BETTER

- ✓ Give practical examples.
- ✓ Talk about real-world use cases.
- ✓ Explain how you investigated incidents.
- ✓ Share KQL queries you have written.
- ✓ Mention automation and playbooks.
- ✓ Show that you think like an analyst, not just an admin.
- ✓ Be clear, concise and confident!



ONE LINE DEFINITIONS (FOR QUICK REVISION)

- Sentinel: Cloud SIEM + SOAR.
- Log Analytics Workspace: Stores logs.
- KQL: Query language for logs.
- Connector: Brings data into Sentinel.
- Analytics Rule: Detects threats.
- Alert: Single detection.
- Incident: Group of related alerts.
- Automation Rule: Triggers playbooks.
- Playbook: Automated response workflow.
- SOAR: Automate & orchestrate response.
- Threat Hunting: Find threats proactively.



Employers don't just want knowledge – they want problem solvers.

Practice. Build. Investigate. Automate. Hunt. Repeat.





25. CAPSTONE PROJECT + CTA



Microsoft
Sentinel

"Now become the analyst."

Build a complete Microsoft Sentinel SOC lab from scratch. Apply everything you learned in this guide and operate like a real SOC analyst!

▶ YOUR 10 STEP CAPSTONE PROJECT ROADMAP



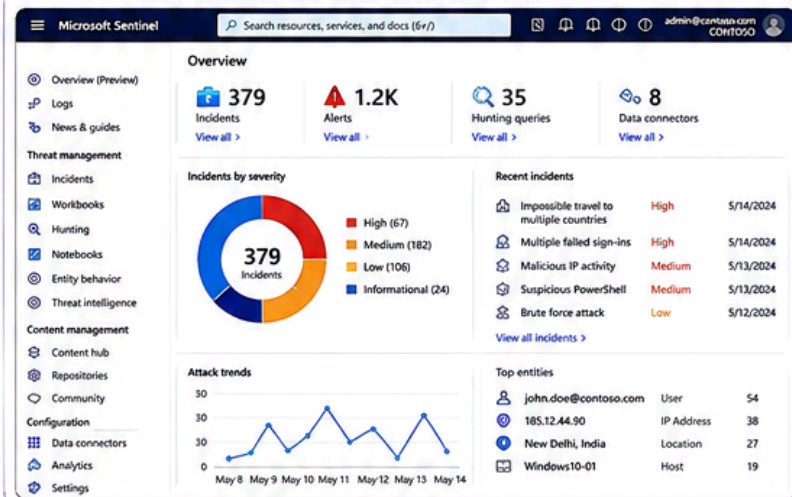
SKILLS YOU WILL MASTER

- ✓ Microsoft Sentinel
 - ✓ KQL Querying
 - ✓ Detection Engineering
 - ✓ Incident Investigation
 - ✓ Threat Hunting
 - ✓ Automation & Playbooks
 - ✓ SOAR (Response)
 - ✓ Dashboards & Reporting
 - ✓ Cloud Security
 - ✓ SOC Analyst Mindset
- You are not just learning a tool, you are building a **career**.

END-TO-END SOC WORKFLOW (WHAT YOU WILL BUILD)



YOUR SENTINEL LAB WILL LOOK LIKE THIS



EXAMPLE DETECTION RULE YOU WILL CREATE

```
// Impossible travel detection rule
let timeWindow = 30m;
let locations = SigninLogs
| where TimeGenerated > ago(1d)
| where ResultType == 0
| summarize first_time = min(TimeGenerated),
last_time = max(TimeGenerated),
locations = make_set(Location, 10)
by UserPrincipalName;
Locations
| where array_length(locations) > 1
and datetime_diff('minute', last_time, first_time) <
timeWindow
```

PLAYBOOK EXAMPLE (Auto Response)



WHAT YOU WILL DELIVER

- ✓ A fully configured Microsoft Sentinel environment
- ✓ Connected data sources (real logs)
- ✓ KQL queries library
- ✓ Working analytics rules
- ✓ Generated alerts and incidents
- ✓ Investigated incidents with evidence
- ✓ Custom workbook dashboard
- ✓ Automated playbook in action
- ✓ Professional incident report

PROJECT OUTCOMES

- ✓ Build real SOC analyst hands-on experience
- ✓ Understand attacker behavior
- ✓ Detect, investigate & respond to threats
- ✓ Automate repetitive tasks
- ✓ Improve detection coverage
- ✓ Create measurable security impact
- ✓ Stand out in interviews & real world

FINAL WORD

"Cloud tools detect threats.
Skilled analysts stop them."

Keep learning. Keep hunting.
Keep protecting.

You've got this! 💪



WANT
MORE?



Guided Labs



Real Screenshots



SOC Projects



Career Roadmaps



Interview Preparation



Mentor Support



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You didn't just read this guide. You built your future.

Stay curious. Stay sharp. Stay blue.

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